



says that Bluespec's abstraction enhanced productivity by at least 10x compared to conventional techniques.

The startup's approach allows for the generation of multiple variants from a single code line, aligning with the fundamental RISC-V principles of flexibility and customisability. The company provides a business model adaptable to clients' requirements through discussions about industry domains, performance requirements, and considerations for real-time systems to understand the unique workloads and roadmaps.

Gala says, "Alongside developing our processor IP portfolio, we are also expanding our operations to become a one-stop shop for the custom silicon needs of our customers. If they choose to leverage our IPs, we will assist them in assembling all the necessary components, crafting a unique solution, and connecting them with partners who can guide you through the manufacturing

process and oversee the rest of the supply chain."

The startup uses standardised interfaces (AHB, APB, AXI) for seamless integration with other IP blocks on a standard SoC. It also offers an interconnect subsystem as part of its IP portfolio, empowering customers to build systems with flexibility and hierarchy. "When you acquire the core from us, you also get the interconnect IP, allowing you to build any system with your chosen hierarchy and bus matrix. You can easily integrate third-party IP for USB, DDR, Ethernet, and more," explains Gala.

InCore's licensing involves negotiable initial non-recurring engineering (NRE) fixed fees and royalties, with flexibility based on factors such as promised volumes or specific customisation needs. The startup believes that its flexibility in tailoring licensing models to suit the dynamic business needs of its partners sets it apart from traditional processor vendors. "The aim is to deliver a customised solution that not only

meets but exceeds the client's initial expectations," says Madhu.

The startup became India's second chip design startup to receive funding from PeakXV Partners (formerly Sequoia Capital India) worth \$3 million, which it will use to enhance its portfolio of Core-Hub generators and reference SoC platforms aimed at multiple segments in the embedded chips space. It is currently looking to partner with a domestic embedded application solutions provider to complement its solutions. "This could include applications like the TCP/IP stack or safety features for automotive solutions, such as low-cost anti-lock braking system (ABS) for motorbikes," says Madhu.

The startup is working to include a third core, comparable with ARM's Cortex-A55, with speeds reaching up to two gigahertz. It is also working to enhance its capabilities in vectors, security, and performance monitoring tools.

—Yashasvini Razdan

3 THIS EV MOTOR WORKS WITHOUT RARE EARTH METALS, SAYS TATA ELXSI

Tata Elxsi's rare earth-free motor's simple, robust design minimises wear. As it reshapes the auto landscape, India's reliance on specific sources dwindles, steering away from China's rare-earth dominance

In a conversation with the Electronics For You team, Tata Elxsi revealed it is developing an electric vehicle motor that, instead of rare-earth metals, relies on ferrite.

The company asserts that this motor will assist India and its vehicle manufacturers in reducing dependence on China for the importation of rare-earth metals. Rare earth metals, such as Neodymium (Nd) and Dysprosium (Dy), find use in permanent magnet motors globally. China dominates the production of Nd and

Dy, contributing to 67% of the total, followed by Myanmar (12%), Australia (10%), and the United States (9%). Conversely, ferrite is produced by combining iron oxide with metallic elements like barium, manganese, nickel, strontium, and zinc. According to the Volza website, India ranks among the top three importers of Dy and Nd.

Moreover, this motor is touted to be economically viable for both production and servicing. Anush G. Nair, Senior Technology Manager,

Electrification Practice Group at Tata Elxsi, indicated that the motor has undergone extensive testing, yielding better results than anticipated.

"Its construction is simple and rugged, resulting in reduced wear and tear. The intricacies lie in control and software, but these should prove manageable in the long run, given their reliance on power electronics and software. India's dependence on a specific geographical location for these components is minimal," he explains.